VINICIUS CHRISOSTHEMOS TEIXEIRA

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EDUCATION

2024 - Present
Porto Alegre - Brazil

PhD degree in Computer Science
Pontifical Catholic University of Rio Grande do Sul (PUCRS)
- Focus on models for generating training for manual tasks in augmented reality using demonstrations.

2022 - 2023
Porto Alegre - Brazil

Master's degree in Computer Science
Pontifical Catholic University of Rio Grande do Sul (PUCRS)
- Focus on using Virtual and Augmented Reality to impove teaching and learning methods

2017 - 2022
Porto Alegre - Brazil

Bachelor's degree in Computer Science
Pontifical Catholic University of Rio Grande do Sul (PUCRS)
- Participating in 3 research groups

2014 - 2015
São Leopoldo - Brazil

Computer Technician
PRONATEC
- Learning basic knowledge in programming, networks, databases and more

EXPERIENCE

2024 - Present
Porto Alegre - Brazil

Online Undergraduate Tutor
Pontifical Catholic University of Rio Grande do Sul (PUCRS)
- Responsible for answering questions and suggesting improvements in the content produced by the instructors.
- Disciplines: Data Science and Artificial Intelligence, Machine Learning, Data Visualization, and Business Intelligence.

2019 - 2022
Porto Alegre - Brazil

Undergraduate Researcher Project (GRV)
Pontifical Catholic University of Rio Grande do Sul (PUCRS)
- Development of projects involving Computer Vision, Image Processing and Virtual and Augmented Reality.

2021 - 2021
São Paulo - Brazil

Service Provider (Developer)
Pontifical Catholic University of Rio Grande do Sul (PUCRS)
- Development of a respiratory monitor based on image processing.

2019 - 2019
Porto Alegre - Brazil

Undergraduate Research Assistant (SMART)
Pontifical Catholic University of Rio Grande do Sul (PUCRS)
- Assist in the development of a natural disaster simulator to train multi-agent systems techniques. Development of features, tests and graphical interface.

2017 - 2018
Porto Alegre - Brazil

Undergraduate Researcher Project (PET-Informática)
Pontifical Catholic University of Rio Grande do Sul (PUCRS)
- Develop and implement projects involving programming to consolidate the concepts learned in the Computer Science course.

SKILLS

Augmented Reality | Virtual Reality | Game Development | Computer Vision | Image Processing | IA

C | C++ | C# | Java | Python

Unity3D | Blender

PATENTS

Augmented Reality Game: BallonsAR

Developed and implemented software projects to reinforce and apply concepts learned throughout the Computer Science progra.An augmented reality game in which the player needs to pop balloons that fall from the sky. The game was made for Android, using Unity3D.

Institution: INPI, Patent Number: BR512022001295-2

Vehicle and Container Monitoring System with Augmented Reality - Desktop

A graphical interface for the vehicle and container monitoring system. The interface updates data obtained by sensors in the environment in real time.

Institution: INPI, Patent Number: BR512022001555-2

Movement Detection, Monitoring and Synchronization System

A system that tracks a tag in a camera-captured image in real time and generates movement data based on the tag's behavior.

Institution: INPI, Patent Number: BR102021015787-9

Vehicle and Container Monitoring System with Augmented Reality - Mobile

A system that monitors the entry and exit of vehicles and updates data from a server. A mobile app consumes the updated data and displays the information in augmented reality.

Institution: INPI, Patent Number: BR512022001559-5

Video-based Breathing Monitor

The software is used to identify a label glued to the abdomen of premature children and generate a graph based on the movement caused by breathing in the child's abdomen. The objective of this software is to provide a non-invasive method for detecting the respiratory cycle of premature babies.

Institution: INPI, Patent Number: BR512022000612-0

PROJECTS

Virtual Museum

This project is a Virtual Reality application developed in Unity, designed to support the learning process of students with special educational needs, such as cerebral palsy, dyscalculia, and low vision. The experience recreates a virtual museum environment, inspired by the PUCRS Museum, where students can explore interactive exhibits in an immersive and engaging way.

Demo: https://youtu.be/-J63ma1XvAU

Virtual Bike

This project is a Virtual Reality application that simulates riding a bicycle using real-world sensors and a 3D environment built in Unity for the Meta Quest 3.

Demo: https://www.youtube.com/watch?v=GAMwGyg6p1Y

Game inspired in Dispatch

This project is a clone of the game Dispatch developed using Unity, created solely for educational and learning purposes in Unity game development. The main goal is to study and replicate the mechanics, systems, and visual elements of the original game in order to improve skills in programming, game design, and Unity tools.

Demo: https://www.youtube.com/watch?v=ebOV1wkTgug

Soccer Game Replay in Virtual Reality

Developed a mobile application that enables users to download football match replays from a remote server and view them in Virtual Reality from multiple predefined perspectives. Implemented features for replay downloading, stadium big-screen playback, and immersive on-field visualization.

Demo: https://www.youtube.com/watch?v=DELOkUuWptk

Builing AR Scenes with Soccer Players

Developed an Augmented Reality application that enables users to place, manipulate, and remove virtual objects in real-world environments. Implemented surface detection and object interaction features, including placement, repositioning, and deletion.

Demo: https://www.youtube.com/watch?v=LYU5auk4eyE

Video respiratory monitor

Developed a mobile application for non-contact respiratory monitoring of premature infants using image processing techniques. The application automatically detects fiducial markers, defines a region of interest (ROI), measures abdominal motion to estimate the respiratory cycle, and records video data for subsequent graphical analysis.

Demo: https://github.com/ViniciusChrisosthemos/Portifolio?tab=readme-ov-file#video-respiratory-monitor

Remote Assistance System in Augmented Reality

Developed a client-server system enabling remote communication between an Oculus Quest headset and a desktop application. The client detects fiducial markers, streams its first-person view to the server, and receives virtual object placements that are rendered in the user's field of view.

Demo: https://www.youtube.com/watch?v=JWfYCmh9_Lc

Simulator data viewer in Virtual Reality

Developed a Virtual Reality application for visualizing crowd simulation data generated by the LODUS simulator. The application enables users to analyze population movement between regions of interest, monitor healthy and infected individuals, and explore crowd behavior throughout the simulation.

Demo: https://www.youtube.com/watch?v=iqSWtQkHDqQ

Simulator data viewer in Virtual Reality

Designed and developed a Virtual and Augmented Reality training platform for industrial machine maintenance. The system enables users to create, edit, and execute interactive maintenance procedures, synchronize training sessions from a remote server, and manage workflows through an integrated control panel. Implemented support for dynamic asset loading via Asset Bundles, allowing users to place and configure virtual objects within the environment.

Demo: https://www.youtube.com/watch?v=klzwiG3Lrao

PUBLICATIONS

(Best Paper in Conference) Development of a Respiratory Motion Video-Based Monitor for Non-Invasive Mechanical Ventilation Synchronization in Premature Infants

2025 IEEE 49th Annual Computers, Software, and Applications Conference (COMPSAC)

Carine L. Rech, Vinicius Chrisosthemos T., Humberto H. Fiori, Márcio S. Pinho

2025 DOI: 10.1109/COMPSAC65507.2025.00112

Accurate respiratory monitoring remains a challenge, which hinders synchronization with mechanical ventilators. This problem is even more pronounced in premature infants. Monitoring must be sensitive enough to detect respiratory movement rapidly and accurately without adding risks to the patient. There is still no sensitive, cheap, and non-invasive device available.This is an experimental study to validate the respiratory motion video monitor equipment, including premature patients on non-invasive ventilation support. A marking tape (tag) is placed between the patient's chest and abdomen. A camera is positioned over the incubator to capture the breathing movement through the tape, and a graph is generated. Another camera records the ventilator reading its trace. The graphs of both videos and the direct visualization of the patient are simultaneously synchronized in the software developed. The beginning and end of breathing and the patient's, ventilator's, and video monitor's respiratory frequency are analyzed and then compared to each other.Regarding respiratory rate, the video monitor proved to be more accurate than the internal flow sensor of the respirator, detecting 93.3% of the patient's respiratory movements, while the respirator detected only 73.2%. In terms of the onset of breathing, the video monitor detected it before direct patient visualization (-0.05s), and was also better compared to the respirator, which detected it after direct patient visualization (0.11s). The respiratory video monitor is relatively accurate, has good precision in reading the onset of patient respiratory effort and respiratory rate.

A Novel AI-driven Automated Orthodontic Model Analysis to Improve Classification of Orthodontic Extraction Cases

2025 IEEE 49th Annual Computers, Software, and Applications Conference (COMPSAC)

Sunna I. Ahmad, Adriel S. Araújo, Vinicius Chrisosthemos T., Carlos F. A. Gomes, Vinicius Dutra, Quinn Roederer, R. Scott Conley, Dalvan Griebler, Márcio S. Pinho, Hakan Turkkahraman

2025 DOI: 10.1109/COMPSAC65507.2025.00254

Malocclusion, a prevalent dental condition worldwide, necessitates orthodontic intervention to correct tooth misalignment and improve oral health. Treatment can involve extraction of permanent teeth, depending on dental crowding, jaw relationships, and facial aesthetics. Today, clinical decision support systems have introduced machine learning (ML) to assist orthodontists in determining optimal treatment plans. This study explores the development of a novel, fully automated method for extracting dentoalveolar features from 3D intraoral scans (IOS), aiming to enhance orthodontic decision-making. Using deep learning-based IOS segmentation as basis, dental measurements were developed and utilized to train supervised ML classifiers, including support vector machines (SVM), logistic regression, decision trees, and random forests. An ensemble of SVM models demonstrated the highest accuracy (73%) in predicting extraction decisions, with these novel domain-specific features proving more informative than traditional dental arch measurements. While we can make further improvements not only in the automated segmentation but also by applying feature selection, the results highlight the potential of AI-driven analysis to streamline orthodontic workflows, reduce manual intervention and improve clinical efficiency.

Using Frameworks to Develop Augmented Reality Training in Industry: A Systematic Review

Intelligent Systems and Applications

Vinicius Chrisosthemos T., Márcio S. Pinho

2025 DOI: 10.1007/978-3-031-99965-9_21

Augmented Reality is being increasingly used in industrial training, helping to improve learning and task performance. In recent years, various approaches have been developed to create training assembly frameworks that are accessible to industry experts. This study aims to identify these frameworks, analyze the most used approaches, their advantages and disadvantages, and understand how they benefit industrial worker training. For this, a systematic review was conducted in the main databases, defining four research questions and applying inclusion and exclusion criteria. In the final stage, an additional Snowballing process was used to expand the results and include relevant articles in the analysis. The findings shows that AR enhances knowledge retention and reduces errors, though challenges like object tracking and hardware limitations remain. The review found a trend in using the Authoring-by-Demonstration approach, where training is created based on demonstrations. This method showed the most promising results, especially with the rapid advancements in artificial intelligence.

Multiview Machine Learning Classification of Tooth Extraction in Orthodontics Using Intraoral Scans

2024 IEEE 48th Annual Computers, Software, and Applications Conference (COMPSAC)

Carlos F. A. Gomes, Adriel S. Araújo, Sunna I. Ahmad, Mauricio C. Magnaguagno, Vinicius Chrisosthemos T., Anushri S. Rajapuri, Quinn Roederer, Dalvan Griebler, Vinicius Dutra, Hakan Turkkahraman, Márcio S. Pinho

2024 DOI: 10.1109/COMPSAC61105.2024.00316

Orthodontic treatment planning often involves de-ciding whether to extract teeth, a critical and irreversible decision. Integrating machine learning (ML) can enhance decision-making. This study proposes using Intraoral Scans (IOS) 3D models to predict extraction/non-extraction binary decisions with ML models. We leverage a multiview approach, using images taken from multiple points of view of the 3D model. The methodology involved a dataset composed of preprocessed IOS from 181 subjects and an experimental procedure that evaluated multiple ML models in their ability to classify subjects using either grayscale pixel intensities or radiomic features. The results indicated that a logistic model applied to the radiomic features from the back and frontal views of the 3D models was one of the best model candidates, achieving a test accuracy of 70 % and F1 score of. 73 and. 65 for non-extraction and extraction cases, respectively. Overall, these findings indicate that a multiview approach to IOS 3D models can be used to predict extraction/non-extraction decisions. In addition, the results suggest that radiomic features provide useful information in the analysis of IOS data.

PUBLICATIONS

Automatic Detection of Common Gastroenterological Diseases Using a Small Dataset: A Two-Phase Image Processing Method

Proceedings of the Future Technologies Conference (FTC) 2024

Rafael N. Copstein, Vicenzo A. Sangalli, Renan M. Trévia, Leonardo R. Amado, Vinicius Chrisosthemos T., Márcio S. Pinho

2024 DOI: 10.1007/978-3-031-73125-9_1

Manually analyzing images from LASER Confocal Endomicroscopy (CLE) images is a common approach for diagnosing gastroenterological diseases that requires training and experience. The subjective aspect of this approach may lead to misdiagnoses, especially when the objective is to identify which of the diseases that can lead to esophageal cancer the patient possess: Gastric Metaplasia, Barret's Esophagus, or Neoplastic Mucosa. Automating the diagnosis has been explored by some authors, but without regard for specific disease identification. This work proposes a classification method for images obtained from CLE exams that combine results of two classifiers based on features obtained from Gray Level Co-occurrence Matrices and Local Binary Patterns, two traditional texture description methods. The approach divides a CLE image into four regions of interest and then classifies each region with both classifiers. The final step combines the results of both classifiers using a consensus algorithm into a final prediction of the original image. The proposed method could classify the images by the disease they represent, not only by whether they contain or not a disease, which is the most common approach by similar methods. The method also achieved a result comparable to related work despite using a much smaller dataset for training.

Visualization of Crowd Contamination Simulations Using Immersive Virtual Reality

2024 IEEE 48th Annual Computers, Software, and Applications Conference (COMPSAC)

Vinicius Chrisosthemos T., Gabriel F. Silva, Isabel H. Manssour, Soraia H. Musse, Márcio S. Pinho

2024 DOI: 10.1109/COMPSAC61105.2024.00287

Smart cities can generate a lot of useful data for analyzing behavior and infrastructures, helping with urban planning. However, the applications found are often limited to desktop viewing only, which can hinder or restrict data analysis. This work proposes an immersive data visualization of the population of medical centers to assist in the analysis and development of new strategies to deal with overcrowding situations. The prototype allows the visualization of crowd data located in places of interest on top of a 3D map. The user can view the number of people present at the location at the current stage of the simulation or over time. To test the prototype, data from an infectious disease simulator was used, called LODUS. The results showed that the application has potential for visualizing this type of data.

Investigating Radiological Diagnosis Through Smartphone-Based Virtual Reality Applications: A User Study

2024 IEEE 48th Annual Computers, Software, and Applications Conference (COMPSAC)

Renan Trévia, Vinicius Chrisosthemos T., Márcio S. Pinho

2024 DOI: 10.1109/COMPSAC61105.2024.00339

This study investigates the utilization of low-cost three-dimensional visualization devices., specifically virtual reality applications on smartphones., to enhance radiological diagnosis processes. Radiology., a critical field in medicine., often faces challenges such as external illumination and poor ergonomic conditions during diagnostic procedures. Virtual reality technology has emerged as a potential solution to address these issues. The research conducted user studies with radiologists and medical physicists to assess the effectiveness and usability of the virtual reality application. Feedback from participants indicated positive perceptions of the virtual environment., 3D model quality., and interaction with the application. The study also evaluated geometric transformations on the 3D model, highlighting the importance of user-friendly controls. Overall, the findings suggest that virtual reality technology on smartphones holds promise in supporting radiological diagnosis by providing an immersive and efficient tool for medical professionals.

A Framework for AR Treasure Hunt Design

SVR '23: Proceedings of the 25th Symposium on Virtual and Augmented Reality

Vinicius Chrisosthemos T., Márcio S. Pinho

2023 DOI: 10.1145/3625008.3625050

This paper proposes a framework for the development of Treasure Hunt games using Augmented Reality for educational purposes. The Framework is based on markers and allows the insertion of clues in the form of a simple instruction or an objective question with multiple choices along with 3D models anchored on the marker. Our main contribution was the development of a method for editing the game directly on the display device, without the need to adjust the content in a desktop editor and with low complexity of use. Overall, the framework was able to create Treasure Hunt applications in an easy and intuitive way.

Immersive Modeling Framework for Training Applications

2023 IEEE International Conference on Advanced Learning Technologies (ICALT)

Vinicius Chrisosthemos T., Carlo S. Mantovani, Alexandre Cardoso, Carlos A. dos Santos, Márcio S. Pinho

2023 DOI: 10.1109/ICALT58122.2023.00087

This article describes a framework for modeling and executing training in augmented and virtual reality environments. The framework was designed based on characteristics observed in existing training applications for complex tasks, such as the use of tools, control panels and the need for step-by-step instructions. Unlike other frameworks, in the proposed system, it is possible to create the training entirely within the virtual/augmented environment, avoiding constant switching between 2D and 3D environments. The framework allows for the definition of steps in a training program, each of which includes textual instructions, videos, and 3D objects, static or animated, anchored in the real world. To demonstrate the capabilities of the framework, a training program for operating a Universal Testing Machine was created as a case study. Overall, the proposed framework allows for the creation of effective and efficient AR training programs for a variety of tasks and industries.